

PHL 1010: Magnetostatics

Many historians of science believe that the compass, which uses a magnetic needle, was used in China as early as the 13th century BC, **its invention being of Indian origin...**

*Raymond A. Serway
John W. Jewett*

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Loadstone



Thales of Miletus

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1819: Current creates magnetic field
...at last determined relation between E & M

HANS CHRISTIAN OERSTED
Danish Physicist and Chemist (1777–1851)



1831: Law of EM induction, Self/mutual inductance, Magneto-optic effect...

MICHAEL FARADAY
British Physicist and Chemist (1791–1867)

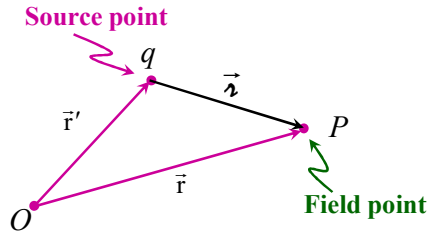


1864: Vacuum polarization current (electric counterpart of Faraday's magnetic induction)
EM Theory (field equations)

JAMES CLERK MAXWELL
Scottish Theoretical Physicist (1831–1879)

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Convention, that would be followed...



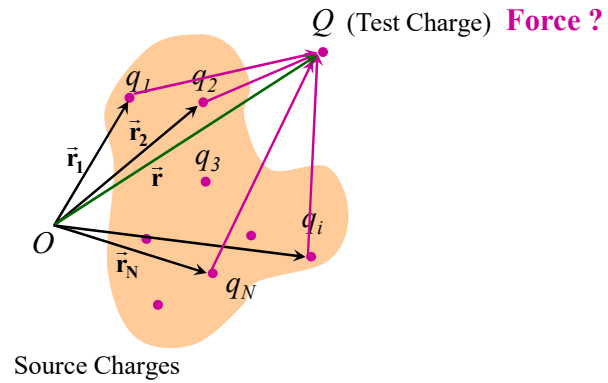
Source point q

Field point P

$\vec{r} = \vec{r} - \vec{r}'$

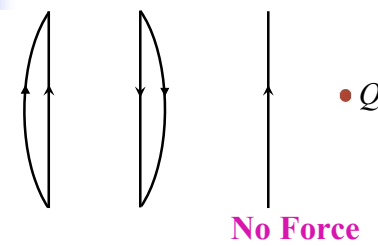
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The Fundamental Problem of EM Theory



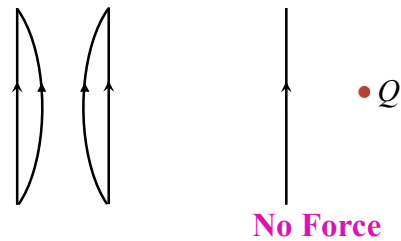
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Consider the simplest & Historical Experiment



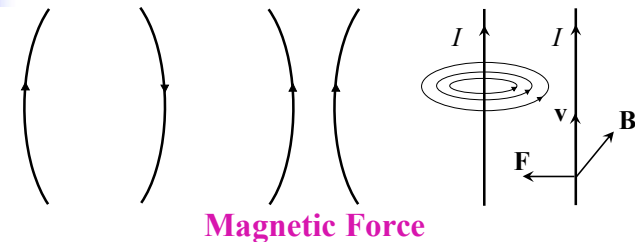
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Consider the simplest & Historical Experiment



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Consider the simplest & Historical Experiment



$$\text{Lorentz Force Law: } \vec{F}_{\text{mag}} = q(\vec{v} \times \vec{B})$$

$$\vec{F}_{\text{tot}} = q[\vec{E} + (\vec{v} \times \vec{B})]$$

Work done by magnetic force: q moves by $d\vec{l} = \vec{v} dt$

$$dW_{\text{mag}} = \vec{F}_{\text{mag}} \cdot d\vec{l} = Q(\vec{v} \times \vec{B}) \cdot \vec{v} dt = 0 \quad \text{Magnetic force does no work}$$

Magnetic Force can never accelerate/decelerate a particle set in motion

[Example 5.3 HW]

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Motion of Charges particle under Magnetic Force

Lorentz Force Law: $\vec{F}_{\text{mag}} = q(\vec{v} \times \vec{B})$

$(y - R\omega t)^2 + (z - R)^2 = R^2$ $\omega = qB/m$; $R = E/\omega B$

[HW]

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Important concepts Current

Current: $I = \frac{dq}{dt}$

Line current

$dI = I dl$ ✓

$I = \frac{dq}{dt} = \frac{\lambda v dt}{dt} = \lambda v$

$\vec{I} = \lambda \vec{v}$

Line current distribution → Line current density

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Important concepts Surface Current

Line element Line element

$dI = \left(\frac{I}{d}\right) dx = K dx$

$K = \frac{dI}{dx}$

$K = \frac{dI}{dl_{\perp}}$

$I = \int \vec{K} \cdot d\vec{l}$

$dI = \frac{dQ}{dt} = \frac{\sigma dl_{\perp} dl_{\parallel}}{dt} = \sigma dl_{\perp} \frac{dl_{\parallel}}{dt} = \sigma dl_{\perp} v$

$\frac{dI}{dl_{\perp}} = \sigma v = K$ $\sigma \vec{v} = \vec{K}$

Surface current distribution → Surface current density

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Important concepts Volume Current

$dI = \left(\frac{I}{\pi R^2}\right) dx dz$

$J = \frac{dI}{dx dz}$

$J = \frac{dI}{da_{\perp}}$ $I = \int \vec{J} \cdot d\vec{a}$

$dI = \frac{dQ}{dt} = \frac{\rho da_{\perp} dy}{dt}$

$= \rho da_{\perp} \frac{dy}{dt} = \rho da_{\perp} v$

$\frac{dI}{da_{\perp}} = \rho v = J$ $\rho \vec{v} = \vec{J}$

Volume current distribution → Volume current density

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Important concepts**Volume Current**

Example: Current density is proportional to the distance from the axis $J = k r$

Find total current.



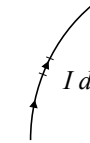
$$I = \int \vec{J} \cdot d\vec{a}$$

$$I = \int (kr) r dr d\phi = 2\pi k \int_0^a r^2 dr = \frac{2\pi k a^3}{3} \quad \checkmark$$

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Important concepts**Summary**

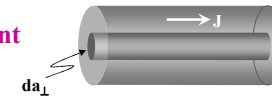
Line current



Surface current



Volume current



$$dI = I dl = K dx = J dx dz$$

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Consequence of Lorentz Force: Force on Current

Magnetic Force on a current element:

$$\vec{F}_{\text{mag}} = \int dq (\vec{v} \times \vec{B}) = \int \lambda dl (\vec{v} \times \vec{B}) = \int dl (\vec{I} \times \vec{B}) = I \int (d\vec{l} \times \vec{B})$$

Magnetic force on surface current density:

$$\vec{F}_{\text{mag}} = \int dq (\vec{v} \times \vec{B}) = \int \sigma da (\vec{v} \times \vec{B}) = \int (\vec{K} \times \vec{B}) da$$

Magnetic force on volume current density:

$$\vec{F}_{\text{mag}} = \int dq (\vec{v} \times \vec{B}) = \int \rho d\tau (\vec{v} \times \vec{B}) = \int (\vec{J} \times \vec{B}) d\tau$$

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Important concepts

$I = \oint \vec{J} \cdot d\vec{a}$ = total charge passing through a closed surface per unit time

$$= \int_V (\vec{\nabla} \cdot \vec{J}) d\tau$$

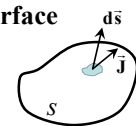
= total charge leaving a volume V per unit time

= rate of which charges are decreasing with time within the closed surface

$$= -\frac{\partial}{\partial t} \int_V \rho d\tau = -\int_V \frac{\partial \rho}{\partial t} d\tau \longrightarrow \int_V \left(\vec{\nabla} \cdot \vec{J} + \frac{\partial \rho}{\partial t} \right) d\tau = 0$$

$$\longrightarrow \vec{\nabla} \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0 \quad \text{Continuity Equation} \\ \text{Conservation of charge}$$

Imp: $\frac{\partial \rho}{\partial t} = 0 \longrightarrow \text{Steady state current} \longrightarrow \vec{\nabla} \cdot \vec{J} = 0$



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Magnetic Field of a Steady Current

Stationary charges $\longrightarrow$ Constant electric field **Electrostatics**

Steady current $\longrightarrow$ Constant magnetic field **Magnetostatics**

$$\frac{\partial \rho}{\partial t} = 0; \quad \vec{\nabla} \cdot \vec{J} = 0$$

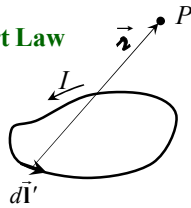
Magnetic field of a steady current: Biot-Savart Law

$$\vec{B}(\mathbf{r}) = \frac{\mu_0 I}{4\pi} \int \frac{d\vec{l}' \times \hat{\mathbf{r}}}{r^2}$$

$$\vec{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{K}(\mathbf{r}') \times \hat{\mathbf{r}}}{r^2} da'$$

$$\vec{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}') \times \hat{\mathbf{r}}}{r^2} d\tau' \quad \mu_0 = 4\pi \times 10^{-7} \text{ N/A}^2$$

1 T = 1 N/A.m
Principle of superposition applicable to magnetic fields

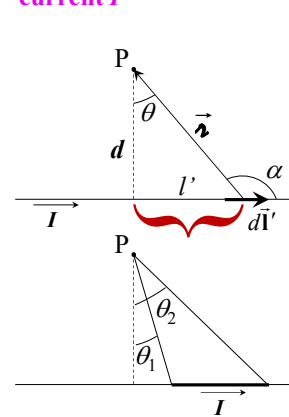


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Magnetic Field of a Steady Current

Magnetic field of a steady current: Biot-Savart Law

Magnetic field at a distance d from a wire carrying steady state current I



$$\vec{B}(\mathbf{r}) = \frac{\mu_0 I}{4\pi} \int \frac{d\vec{l}' \times \hat{\mathbf{r}}}{r^2} \quad 90^\circ - \theta + \alpha = 180^\circ$$

$$|d\vec{l}' \times \hat{\mathbf{r}}| = dl' \sin \alpha = dl' \cos \theta$$

$$\tan \theta = \frac{l'}{d} \longrightarrow dl' = d \sec^2 \theta d\theta$$

$$\cos \theta = \frac{d}{r} \longrightarrow \frac{1}{r^2} = \frac{\cos^2 \theta}{d^2}$$

$$B = \frac{\mu_0 I}{4\pi} \int_{\theta_1}^{\theta_2} d \sec^2 \theta d\theta \cos \theta \frac{\cos^2 \theta}{d^2}$$

$$= \frac{\mu_0 I}{4\pi d} \int_{\theta_1}^{\theta_2} d\theta \cos \theta = \frac{\mu_0 I}{4\pi d} (\sin \theta_2 - \sin \theta_1)$$

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Magnetic Field of a Steady Current

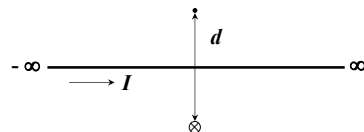
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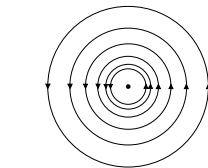
$$B = \frac{\mu_0 I}{4\pi d} (\sin \theta_2 - \sin \theta_1)$$

In case of infinite current carrying wire

$$\theta_2 \rightarrow 90^\circ; \theta_1 \rightarrow -90^\circ$$



$$B = \frac{\mu_0 I}{2\pi d}$$



$$\vec{B} = \frac{\mu_0 I}{2\pi d} \hat{\phi}$$

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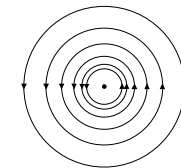
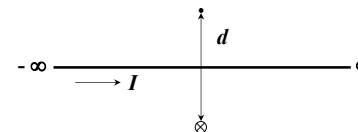
Magnetic Field of a Steady Current

Magnetic field of a steady current: Biot-Savart Law

Magnetic field at a distance d from a wire carrying steady state current I

$$B = \frac{\mu_0 I}{4\pi d} (\sin \theta_2 - \sin \theta_1)$$

In case of infinite current carrying wire



Magnetic field has finite curl !
something you will never see in an electrostatic field

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Divergence & Curl of Magnetic Field

Consider straight current carrying wire

$$\vec{B} = \frac{\mu_0 I}{2\pi d} \hat{\phi}$$

$$d\vec{l} = dr \hat{r} + r d\phi \hat{\phi} + dz \hat{z}$$

$$\oint \vec{B} \cdot d\vec{l} = \frac{\mu_0 I}{2\pi} \oint \frac{1}{r} r d\phi = \frac{\mu_0 I}{2\pi} \int_0^{2\pi} d\phi = \mu_0 I$$

If loop doesn't enclose the wire,

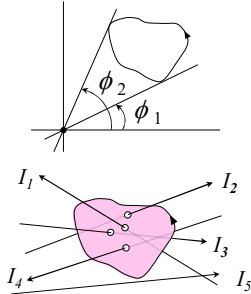
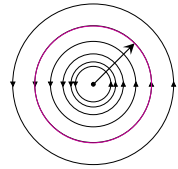
$$\int d\phi = 0 \rightarrow \oint \vec{B} \cdot d\vec{l} = 0 \rightarrow \boxed{\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{en}}$$

$$\rightarrow \oint \vec{B} \cdot d\vec{l} = \mu_0 \sum_{i=1}^4 I_i$$

$$\rightarrow \oint \vec{B} \cdot d\vec{l} = \mu_0 I_{en} = \mu_0 \int \vec{J} \cdot d\vec{a}$$

$$\rightarrow \int (\vec{\nabla} \times \vec{B}) \cdot d\vec{a} = \mu_0 \int \vec{J} \cdot d\vec{a}$$

$$\rightarrow \boxed{\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}} \quad \text{Limitation: Restriction to infinite current}$$



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Divergence & Curl of Magnetic Field

General form of Biot-Savart law :

$$\vec{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{J}(\mathbf{r}') \times \hat{\mathbf{r}}}{r^2} d\tau'$$

$$\vec{B} = \vec{B}(x, y, z)$$

$$\vec{J} = \vec{J}(x', y', z')$$

$$r^2 = (x - x')^2 + (y - y')^2 + (z - z')^2$$

$$d\tau' = dx' dy' dz'$$

Integration is over primed coordinates. Let's take div and curl w. r. t. unprimed coordinates.

$$\vec{\nabla} \cdot \vec{B} = \frac{\mu_0}{4\pi} \int \vec{\nabla} \cdot \left(\vec{J} \times \frac{\hat{\mathbf{r}}}{r^2} \right) d\tau'$$

$$\vec{\nabla} \cdot \left(\vec{J} \times \frac{\hat{\mathbf{r}}}{r^2} \right) = \frac{\hat{\mathbf{r}}}{r^2} \cdot (\vec{\nabla} \times \vec{J}) - \vec{J} \cdot \left(\vec{\nabla} \times \frac{\hat{\mathbf{r}}}{r^2} \right) = 0 \rightarrow \boxed{\vec{\nabla} \cdot \vec{B} = 0}$$

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Divergence & Curl of Magnetic Field

General form of Biot-Savart law :

$$\vec{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{J}(\mathbf{r}') \times \hat{\mathbf{r}}}{r^2} d\tau'$$

$$\vec{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{J}(\mathbf{r}') \times \hat{\mathbf{r}}}{r^3} d\tau'$$

But $\frac{\hat{\mathbf{r}}}{r^3} = -\vec{\nabla} \left(\frac{1}{r} \right)$

$$\vec{\nabla} \times (f\vec{A}) = f \vec{\nabla} \times \vec{A} - \vec{A} \times \vec{\nabla} f$$

$$\vec{B}(\mathbf{r}) = -\frac{\mu_0}{4\pi} \int \vec{J}(\mathbf{r}') \times \vec{\nabla} \left(\frac{1}{r} \right) d\tau'$$

$$\vec{\nabla} \times \frac{\vec{J}(\mathbf{r}')}{r} = \frac{1}{r} \vec{\nabla} \times \vec{J}(\mathbf{r}') - \vec{J}(\mathbf{r}') \times \vec{\nabla} \frac{1}{r}$$

$$\therefore \vec{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \vec{\nabla} \times \frac{\vec{J}(\mathbf{r}')}{r} d\tau' \quad \boxed{\vec{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \vec{\nabla} \times \int \frac{\vec{J}(\mathbf{r}')}{r} d\tau'}$$

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Divergence & Curl of Magnetic Field

$$\therefore \vec{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \vec{\nabla} \times \int \frac{\vec{J}(\mathbf{r}')}{r} d\tau'$$

Taking curl $\vec{\nabla} \times \vec{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \vec{\nabla} \times \vec{\nabla} \times \int \frac{\vec{J}(\mathbf{r}')}{r} d\tau'$

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{A}) = \vec{\nabla} (\vec{\nabla} \cdot \vec{A}) - \nabla^2 \vec{A} \quad \vec{\nabla} \cdot (f\vec{A}) = f \vec{\nabla} \cdot \vec{A} + \vec{\nabla} f \cdot \vec{A}$$

$$\vec{\nabla} \times \vec{\nabla} \times \int \frac{\vec{J}(\mathbf{r}')}{r} d\tau' = \vec{\nabla} (\vec{\nabla} \cdot \int \frac{\vec{J}(\mathbf{r}')}{r} d\tau') - \nabla^2 \int \frac{\vec{J}(\mathbf{r}')}{r} d\tau'$$

$$= \vec{\nabla} \left(\int \vec{\nabla} \cdot \frac{\vec{J}(\mathbf{r}')}{r} d\tau' \right) - \int \nabla^2 \frac{\vec{J}(\mathbf{r}')}{r} d\tau'$$

$$= \vec{\nabla} \int \left[\frac{1}{r} \vec{\nabla} \cdot \vec{J}(\mathbf{r}') + \vec{J}(\mathbf{r}') \cdot \vec{\nabla} \left(\frac{1}{r} \right) \right] d\tau' - \int \vec{J}(\mathbf{r}') \nabla^2 \left(\frac{1}{r} \right) d\tau'$$

$$\vec{\nabla} \times \vec{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \vec{\nabla} \int \vec{J}(\mathbf{r}') \cdot \vec{\nabla} \left(\frac{1}{r} \right) d\tau' - \frac{\mu_0}{4\pi} \int \vec{J}(\mathbf{r}') \nabla^2 \left(\frac{1}{r} \right) d\tau'$$

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Divergence & Curl of Magnetic Field

$$\vec{\nabla} \times \vec{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \vec{\nabla} \int \vec{J}(\mathbf{r}') \cdot \vec{\nabla} \left(\frac{1}{r} \right) d\tau' - \frac{\mu_0}{4\pi} \int \vec{J}(\mathbf{r}') \nabla^2 \left(\frac{1}{r} \right) d\tau'$$

$$\vec{\nabla} \left(\frac{1}{r} \right) = -\vec{\nabla}' \left(\frac{1}{r} \right) \quad \nabla^2 \left(\frac{1}{r} \right) = -4\pi\delta^3(\vec{r})$$

$$\vec{\nabla} \times \vec{B}(\mathbf{r}) = -\frac{\mu_0}{4\pi} \vec{\nabla} \int \vec{J}(\mathbf{r}') \cdot \vec{\nabla}' \left(\frac{1}{r} \right) d\tau' + \frac{\mu_0}{4\pi} \int \vec{J}(\mathbf{r}') 4\pi\delta^3(\vec{r}) d\tau'$$

Integration by part

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Important Tool: Integration by Part

$$\vec{\nabla} \cdot (f\vec{A}) = f(\vec{\nabla} \cdot \vec{A}) + (\vec{\nabla}f) \cdot \vec{A}$$

$$\int_{vol} \vec{\nabla} \cdot (f\vec{A}) d\tau = \int_{vol} f(\vec{\nabla} \cdot \vec{A}) d\tau + \int_{vol} (\vec{\nabla}f) \cdot \vec{A} d\tau = \oint_S f\vec{A} \cdot d\vec{a}$$

$$\int_{vol} f(\vec{\nabla} \cdot \vec{A}) d\tau = - \int_{vol} (\vec{\nabla}f) \cdot \vec{A} d\tau + \oint_S f\vec{A} \cdot d\vec{a}$$

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Divergence & Curl of Magnetic Field

$$\vec{\nabla} \times \vec{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \vec{\nabla} \int \vec{J}(\mathbf{r}') \cdot \vec{\nabla} \left(\frac{1}{r} \right) d\tau' - \frac{\mu_0}{4\pi} \int \vec{J}(\mathbf{r}') \nabla^2 \left(\frac{1}{r} \right) d\tau'$$

$$\vec{\nabla} \left(\frac{1}{r} \right) = -\vec{\nabla}' \left(\frac{1}{r} \right) \quad \nabla^2 \left(\frac{1}{r} \right) = -4\pi\delta^3(\vec{r})$$

$$\vec{\nabla} \times \vec{B}(\mathbf{r}) = -\frac{\mu_0}{4\pi} \vec{\nabla} \int \vec{J}(\mathbf{r}') \cdot \vec{\nabla}' \left(\frac{1}{r} \right) d\tau' + \frac{\mu_0}{4\pi} \int \vec{J}(\mathbf{r}') 4\pi\delta^3(\vec{r}) d\tau'$$

Integration by part yields

$$\vec{\nabla} \times \vec{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \vec{\nabla} \int \left(\frac{\vec{\nabla}' \cdot \vec{J}(\mathbf{r}')}{r} \right) d\tau' - \frac{\mu_0}{4\pi} \oint_S \frac{\vec{J}(\mathbf{r}')}{r} da' + \mu_0 \vec{J}(\mathbf{r})$$

$$\vec{\nabla} \times \vec{B}(\mathbf{r}) = \mu_0 \vec{J}(\mathbf{r}) \quad \text{Ampere's Law}$$

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Ampere's Law

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}$$

$$\int_S (\vec{\nabla} \times \vec{B}) \cdot d\vec{a} = \mu_0 \int_S \vec{J} \cdot d\vec{a}$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{en}$$

Right hand rule for direction of integration and + ve current

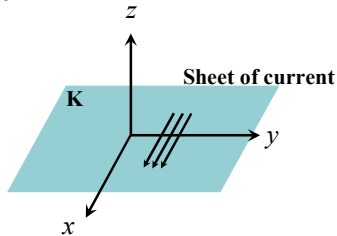
Electrostatics	→	Coulomb Law	Gauss's Law
		↓	↓
Magnetostatics	→	Biot Savart Law	Ampere's Law

Application demand
high degree of symmetry

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Ampere's Law – Applications (2)

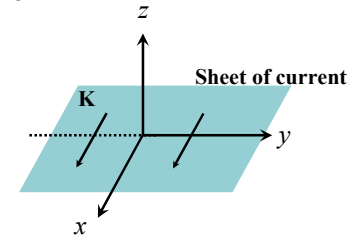
Field of an infinite uniform surface current $\mathbf{K} = K \hat{x}$ flowing over xy-plane



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Ampere's Law – Applications (2)

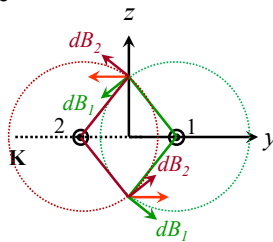
Field of an infinite uniform surface current $\mathbf{K} = K \hat{x}$ flowing over xy-plane



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Ampere's Law – Applications (2)

Field of an infinite uniform surface current $\mathbf{K} = K \hat{x}$ flowing over xy-plane

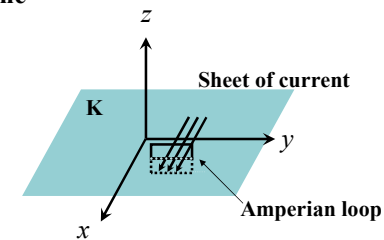


→ Field above is towards left, and field below is towards right

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Ampere's Law – Applications (2)

Field of an infinite uniform surface current $\mathbf{K} = K \hat{x}$ flowing over xy-plane



$$\oint \vec{B} \cdot d\vec{l} = 2Bl = \mu_0 I_{en} = \mu_0 K l$$

$$\vec{B} = -\mu_0 K \hat{y} / 2 \quad z > 0$$

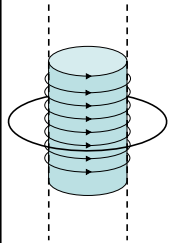
$$\vec{B} = \mu_0 K \hat{y} / 2 \quad z < 0$$

[HW: Examples: 5.9 & 5.10]

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Ampere's Law – Applications (3)

Field of very long solenoid (n closely wound turn per unit length)



Could field have radial component ?

NO

Could field have circumferential component ?

NO

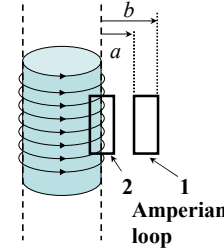
$$\oint \vec{B} \cdot d\vec{l} = B_\phi (2\pi r) = \mu_0 I_{en} = 0$$

→ Field runs parallel to the axis!

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Ampere's Law – Applications (3)

Field of very long solenoid (n closely wound turn per unit length)



Field runs parallel to the axis!

Loop 1:

$$\oint \vec{B} \cdot d\vec{l} = [B(a) - B(b)]L = \mu_0 I_{en} = 0$$

$$B(a) = B(b)$$

Loop 2:

$$\oint \vec{B} \cdot d\vec{l} = BL = \mu_0 I_{en} = \mu_0 n I L$$

$$\vec{B}_{in} = \mu_0 n I \hat{z}$$

$$\vec{B}_{out} = 0$$

HE: Example 5.10

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So far.....

Maxwell Equation for Electrostatics & Magnetostatics

$$\vec{\nabla} \cdot \vec{E} = \frac{1}{\epsilon_0} \sigma \quad \text{Gauss's law}$$

$$\vec{\nabla} \times \vec{E} = 0 \quad \text{No name}$$

$$\vec{\nabla} \cdot \vec{B} = 0 \quad \text{No name}$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} \quad \text{Ampere's Law}$$

→ Magnetic monopole does not exist!

Electric forces are enormously larger than magnetic one

Velocity of source & test charge → Velocity of light

Magnetic field strength → Electric field strength

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Magnetic Vector Potential

In Magnetostatics $\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}$ $\vec{\nabla} \cdot \vec{B} = 0$

In Electrostatics $\vec{\nabla} \times \vec{E} = 0$ → $\vec{E} = -\vec{\nabla} V$

Calculation of Electric Field Calculation of Electric Potential

$\vec{\nabla} \times \vec{B} \neq 0$ → $\vec{B} \neq -\vec{\nabla} \phi$

Strategy $\vec{\nabla} \cdot \vec{B} = 0$ → $\vec{B}$ can be expressed as a curl of a vector (magnetostatic) potential

$$\vec{B} = \vec{\nabla} \times \vec{A}$$

$$\vec{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \vec{\nabla} \times \int \frac{\vec{J}(\mathbf{r}')}{r} d\tau' \quad \rightarrow \quad \vec{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{J}(\mathbf{r}')}{r} d\tau'$$

$$\vec{\nabla} \times \vec{B} = \vec{\nabla} \times \vec{\nabla} \times \vec{A} = \vec{\nabla}(\vec{\nabla} \cdot \vec{A}) - \nabla^2 \vec{A} = \mu_0 \vec{J}$$

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Magnetic Vector Potential	$\vec{B} = \vec{\nabla} \times \vec{A}$
Point 1 A is not uniquely defined $\vec{A} \rightarrow \vec{A}' = \vec{A} + \vec{\nabla} \psi$	
Gauge Transformation	
Point 2 It is always possible to make vector potential divergence-less	
How ?	

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Magnetic Vector Potential	$\vec{B} = \vec{\nabla} \times \vec{A}$
Point 1 A is not uniquely defined $\vec{A} \rightarrow \vec{A}' = \vec{A} + \vec{\nabla} \psi$	
Gauge Transformation	
Point 2 It is always possible to make vector potential divergence-less	
Assume: $\vec{\nabla} \cdot \vec{A} = \lambda$ $\vec{A} \rightarrow \vec{A}' = \vec{A} + \vec{\nabla} \psi$	
$\vec{\nabla} \cdot \vec{A}' = \vec{\nabla} \cdot \vec{A} + \nabla^2 \psi = \lambda + \nabla^2 \psi$	
We can always choose ψ such that $\nabla^2 \psi = -\lambda$	
Poisson's Eq. $\nabla^2 V = -\frac{\rho}{\epsilon_0}$	
$\rho \rightarrow 0$ as $r \rightarrow \infty$ $V = \frac{1}{4\pi\epsilon_0} \int \frac{\rho}{r} d\tau'$	
$\lambda \rightarrow 0$ as $r \rightarrow \infty$ $\psi = \frac{1}{4\pi} \int \frac{\lambda}{r} d\tau'$	

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Magnetic Vector Potential	$\vec{B} = \vec{\nabla} \times \vec{A}$
Point 1 A is not uniquely defined $\vec{A} \rightarrow \vec{A}' = \vec{A} + \vec{\nabla} \psi$	
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We can always choose ψ such that $\nabla^2 \psi = -\lambda$	
$\vec{\nabla} \cdot \vec{A}' = 0$	
$\vec{\nabla} \times \vec{B} = \vec{\nabla} \times \vec{\nabla} \times \vec{A} = \vec{\nabla}(\vec{\nabla} \cdot \vec{A}) - \nabla^2 \vec{A} = \mu_0 \vec{J}$	
$\nabla^2 \vec{A} = -\mu_0 \vec{J}$ Poisson Equation	
$\nabla^2 A_x = -\mu_0 J_x$ $\nabla^2 A_y = -\mu_0 J_y$ $\nabla^2 A_z = -\mu_0 J_z$	

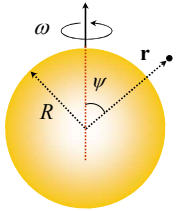
39

Lets conclude...
$\vec{B} = \vec{\nabla} \times \vec{A}$
Gauge Transformation $\vec{A} \rightarrow \vec{A}' = \vec{A} + \vec{\nabla} \psi$
$\vec{\nabla} \cdot \vec{A} = 0$
Poisson Equation $\nabla^2 \vec{A} = -\mu_0 \vec{J}$
$\vec{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}')}{r} d\tau'$
$\vec{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{I}}{r} dl' = \frac{\mu_0}{4\pi} \int I \frac{d\mathbf{l}'}{r}$
$\vec{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{K}}{r} da'$
Calculation of magnetic Field $\rightarrow$ Calculation of magnetic Potential

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Example - 1

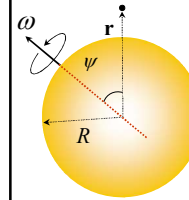
Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



41

Example - 1

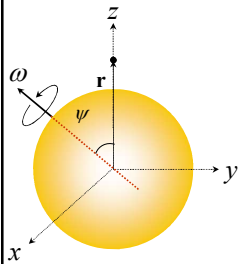
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42

Example - 1

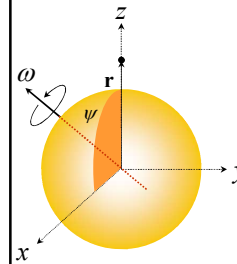
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43

Example - 1

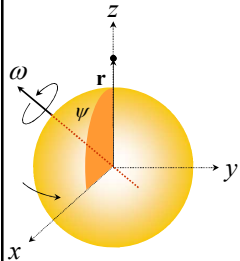
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44

Example - 1

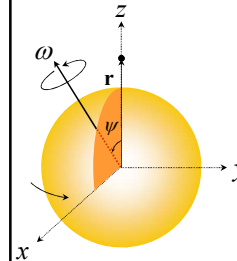
Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



45

Example - 1

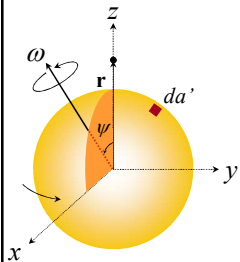
Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



46

Example - 1

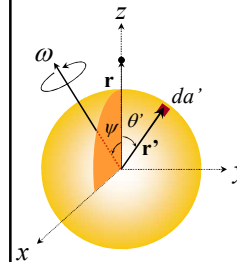
Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



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Example - 1

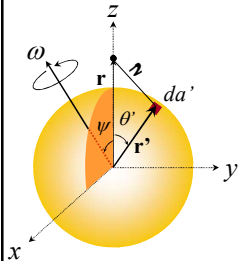
Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



48

Example - 1

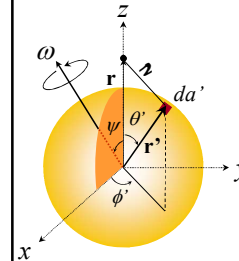
Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



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Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?

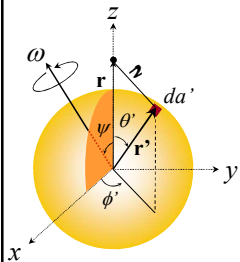


50

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?

Surface charge $\longrightarrow$ Surface current



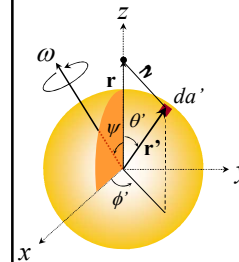
51

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?

Surface charge $\longrightarrow$ Surface current

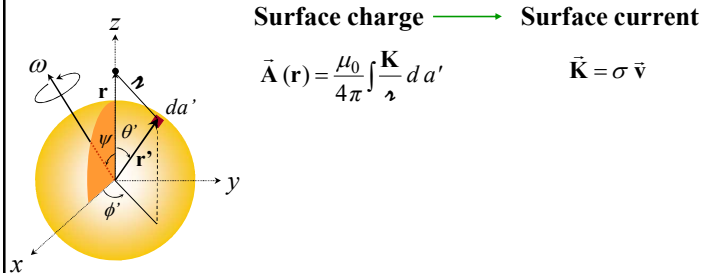
$$\vec{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{K}}{r} da'$$



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Example - 1

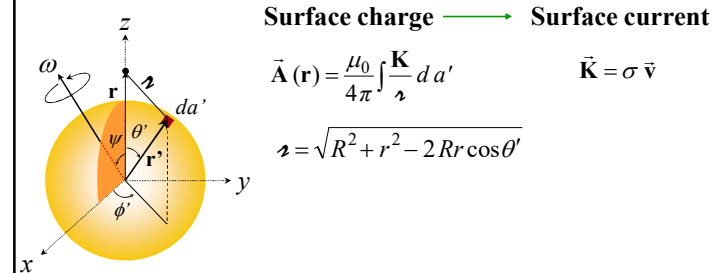
Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



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Example - 1

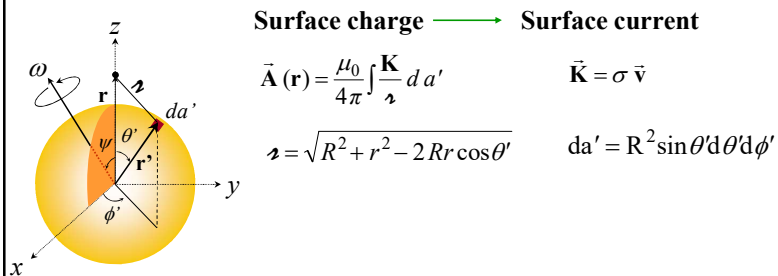
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54

Example - 1

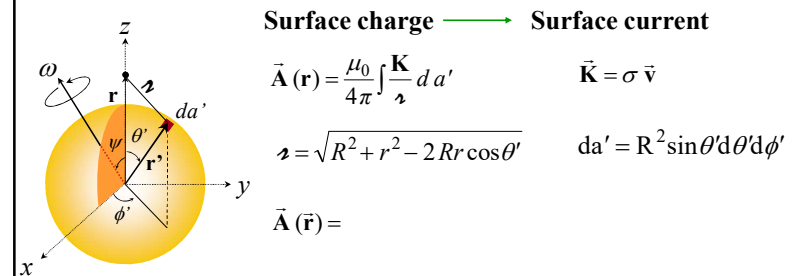
Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



55

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?

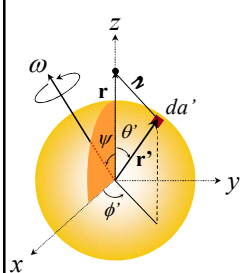


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Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?

Surface charge $\longrightarrow$ Surface current



$$\vec{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{K}}{z} da'$$

$$z = \sqrt{R^2 + r^2 - 2Rr \cos \theta'}$$

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int_0^{2\pi} \int_0^\pi \frac{\sigma \vec{\omega} \times \vec{r}'}{\sqrt{R^2 + r^2 - 2Rr \cos \theta'}} R^2 \sin \theta' d\theta' d\phi'$$

$$\vec{K} = \sigma \vec{v}$$

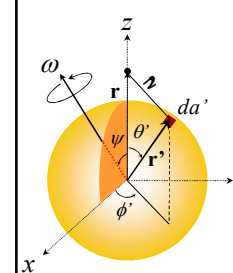
$$da' = R^2 \sin \theta' d\theta' d\phi'$$

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Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?

Surface charge $\longrightarrow$ Surface current



$$\vec{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{K}}{z} da'$$

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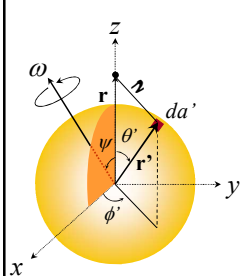
$$da' = R^2 \sin \theta' d\theta' d\phi'$$

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Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?

Surface charge $\longrightarrow$ Surface current



$$\vec{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{K}}{z} da'$$

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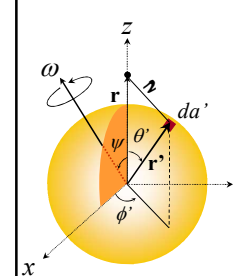
$$da' = R^2 \sin \theta' d\theta' d\phi'$$

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Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?

Surface charge $\longrightarrow$ Surface current



$$\vec{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{K}}{z} da'$$

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$$\vec{K} = \sigma \vec{v}$$

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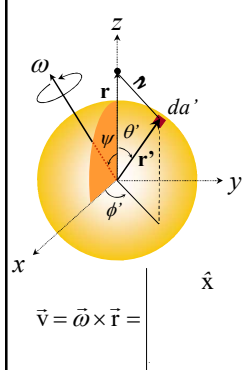
$\vec{v} = \vec{\omega} \times \vec{r}'$

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Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?

Surface charge $\longrightarrow$ Surface current



$$\vec{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{K}}{z} da'$$

$$\vec{K} = \sigma \vec{v}$$

$$z = \sqrt{R^2 + r^2 - 2Rr \cos \theta'}$$

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$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int_0^{2\pi} \int_0^\pi \frac{\sigma \vec{\omega} \times \vec{r}'}{\sqrt{R^2 + r^2 - 2Rr \cos \theta'}} R^2 \sin \theta' d\theta' d\phi'$$

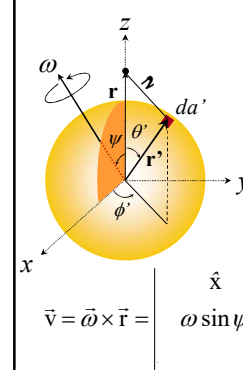
$$\vec{v} = \vec{\omega} \times \vec{r} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \omega \sin \psi & 0 & \omega \cos \psi \end{vmatrix}$$

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Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?

Surface charge $\longrightarrow$ Surface current



$$\vec{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{K}}{z} da'$$

$$\vec{K} = \sigma \vec{v}$$

$$z = \sqrt{R^2 + r^2 - 2Rr \cos \theta'}$$

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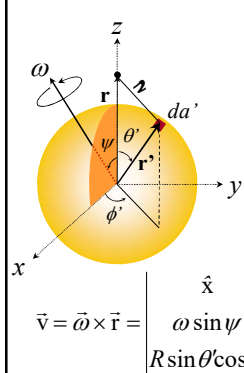
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Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?

Surface charge $\longrightarrow$ Surface current



$$\vec{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{K}}{z} da'$$

$$\vec{K} = \sigma \vec{v}$$

$$z = \sqrt{R^2 + r^2 - 2Rr \cos \theta'}$$

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$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int_0^{2\pi} \int_0^\pi \frac{\sigma \vec{\omega} \times \vec{r}'}{\sqrt{R^2 + r^2 - 2Rr \cos \theta'}} R^2 \sin \theta' d\theta' d\phi'$$

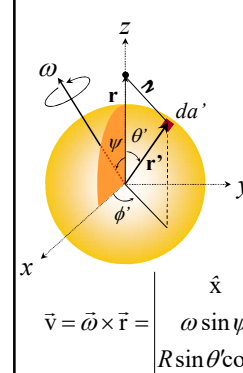
$$\vec{v} = \vec{\omega} \times \vec{r} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \omega \sin \psi & 0 & \omega \cos \psi \\ R \sin \theta' \cos \phi' & R \sin \theta' \sin \phi' & R \cos \theta' \end{vmatrix}$$

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Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?

Surface charge $\longrightarrow$ Surface current



$$\vec{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{K}}{z} da'$$

$$\vec{K} = \sigma \vec{v}$$

$$z = \sqrt{R^2 + r^2 - 2Rr \cos \theta'}$$

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$$\vec{v} = \vec{\omega} \times \vec{r} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \omega \sin \psi & 0 & \omega \cos \psi \\ R \sin \theta' \cos \phi' & R \sin \theta' \sin \phi' & R \cos \theta' \end{vmatrix}$$

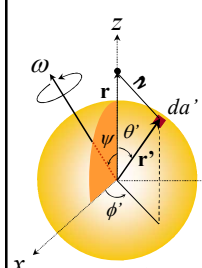
$$= R\omega [-(\cos \psi \sin \theta' \sin \phi') \hat{x}]$$

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Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?

Surface charge $\rightarrow$ Surface current



$$\vec{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{K}}{z} da'$$

$$z = \sqrt{R^2 + r^2 - 2Rr \cos \theta'}$$

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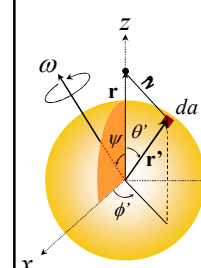
$$= R\omega [-(\cos \psi \sin \theta' \sin \phi')\hat{x} + (\cos \psi \sin \theta' \cos \phi' - \sin \psi \cos \theta')\hat{y} + (\sin \psi \sin \theta' \sin \phi')\hat{z}]$$

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Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?

Surface charge $\rightarrow$ Surface current



$$\vec{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{K}}{z} da'$$

$$z = \sqrt{R^2 + r^2 - 2Rr \cos \theta'}$$

$$da' = R^2 \sin \theta' d\theta' d\phi'$$

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int_0^{2\pi} \int_0^\pi \frac{\sigma \vec{\omega} \times \vec{r}'}{\sqrt{R^2 + r^2 - 2Rr \cos \theta'}} R^2 \sin \theta' d\theta' d\phi'$$

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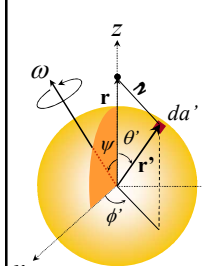
$$= R\omega [-(\cos \psi \sin \theta' \sin \phi')\hat{x} + (\cos \psi \sin \theta' \cos \phi' - \sin \psi \cos \theta')\hat{y} + (\sin \psi \sin \theta' \sin \phi')\hat{z}]$$

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Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?

Surface charge $\rightarrow$ Surface current



$$\vec{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{K}}{z} da'$$

$$z = \sqrt{R^2 + r^2 - 2Rr \cos \theta'}$$

$$da' = R^2 \sin \theta' d\theta' d\phi'$$

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int_0^{2\pi} \int_0^\pi \frac{\sigma \vec{\omega} \times \vec{r}'}{\sqrt{R^2 + r^2 - 2Rr \cos \theta'}} R^2 \sin \theta' d\theta' d\phi'$$

$$\vec{v} = \vec{\omega} \times \vec{r} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \omega \sin \psi & 0 & \omega \cos \psi \\ R \sin \theta' \cos \phi' & R \sin \theta' \sin \phi' & R \cos \theta' \end{vmatrix}$$

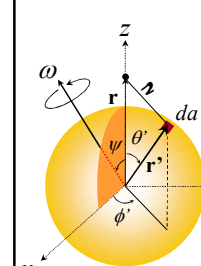
$$= R\omega [-(\cos \psi \sin \theta' \sin \phi')\hat{x} + (\cos \psi \sin \theta' \cos \phi' - \sin \psi \cos \theta')\hat{y} + (\sin \psi \sin \theta' \sin \phi')\hat{z}]$$

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Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?

Surface charge $\rightarrow$ Surface current



$$\vec{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{K}}{z} da'$$

$$z = \sqrt{R^2 + r^2 - 2Rr \cos \theta'}$$

$$da' = R^2 \sin \theta' d\theta' d\phi'$$

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int_0^{2\pi} \int_0^\pi \frac{\sigma \vec{\omega} \times \vec{r}'}{\sqrt{R^2 + r^2 - 2Rr \cos \theta'}} R^2 \sin \theta' d\theta' d\phi'$$

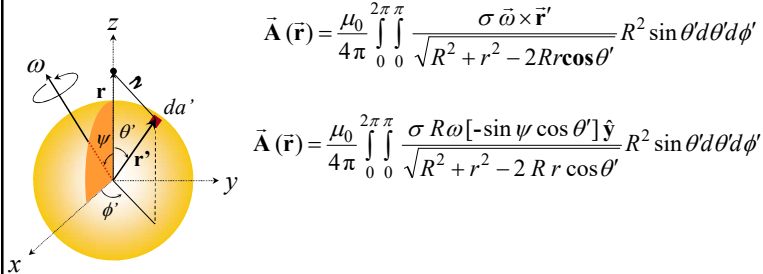
$$\vec{v} = \vec{\omega} \times \vec{r} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \omega \sin \psi & 0 & \omega \cos \psi \\ R \sin \theta' \cos \phi' & R \sin \theta' \sin \phi' & R \cos \theta' \end{vmatrix}$$

$$= R\omega [-(\cos \psi \sin \theta' \sin \phi')\hat{x} + (\cos \psi \sin \theta' \cos \phi' - \sin \psi \cos \theta')\hat{y} + (\sin \psi \sin \theta' \sin \phi')\hat{z}]$$

68

Example - 1

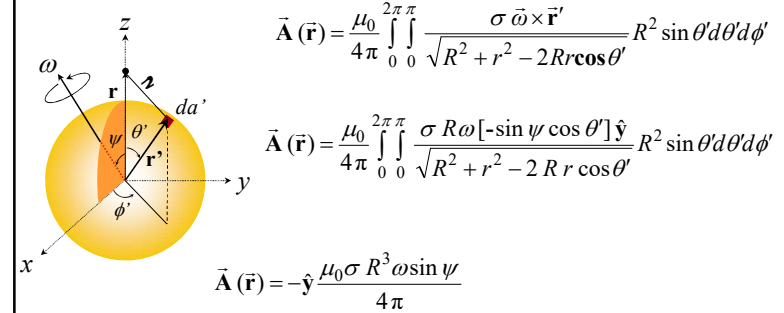
Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



69

Example - 1

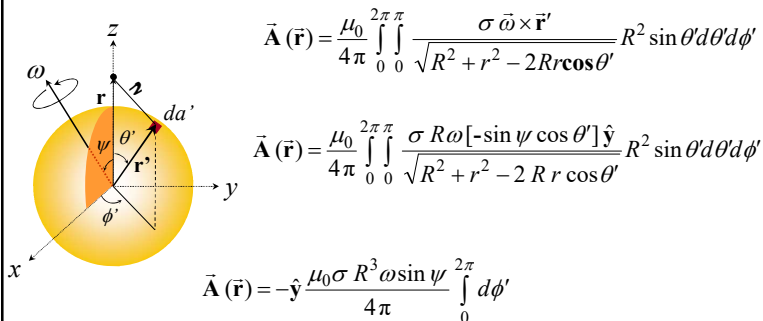
Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



70

Example - 1

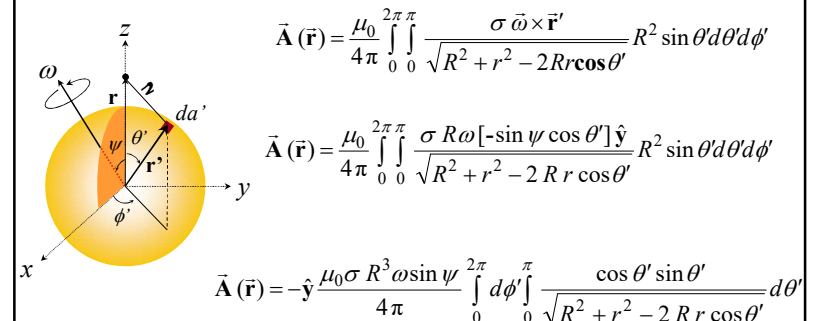
Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



71

Example - 1

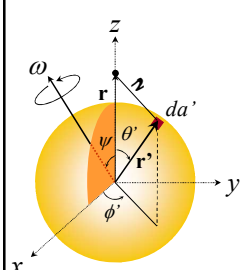
Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



72

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int_0^{2\pi} \int_0^\pi \frac{\sigma \vec{\omega} \times \vec{r}'}{\sqrt{R^2 + r^2 - 2Rr \cos \theta'}} R^2 \sin \theta' d\theta' d\phi'$$

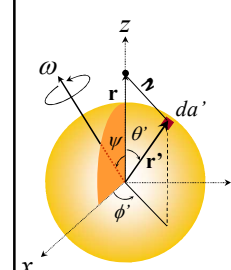
$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int_0^{2\pi} \int_0^\pi \frac{\sigma R \omega [-\sin \psi \cos \theta'] \hat{y}}{\sqrt{R^2 + r^2 - 2Rr \cos \theta'}} R^2 \sin \theta' d\theta' d\phi'$$

$$\vec{A}(\vec{r}) = -\hat{y} \frac{\mu_0 \sigma R^3 \omega \sin \psi}{2 \cdot 4\pi} \int_0^{2\pi} d\phi' \int_0^\pi \frac{\cos \theta' \sin \theta'}{\sqrt{R^2 + r^2 - 2Rr \cos \theta'}} d\theta'$$

73

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int_0^{2\pi} \int_0^\pi \frac{\sigma \vec{\omega} \times \vec{r}'}{\sqrt{R^2 + r^2 - 2Rr \cos \theta'}} R^2 \sin \theta' d\theta' d\phi'$$

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int_0^{2\pi} \int_0^\pi \frac{\sigma R \omega [-\sin \psi \cos \theta'] \hat{y}}{\sqrt{R^2 + r^2 - 2Rr \cos \theta'}} R^2 \sin \theta' d\theta' d\phi'$$

$$\vec{A}(\vec{r}) = -\hat{y} \frac{\mu_0 \sigma R^3 \omega \sin \psi}{2 \cdot 4\pi} \int_0^{2\pi} d\phi' \int_0^\pi \frac{\cos \theta' \sin \theta'}{\sqrt{R^2 + r^2 - 2Rr \cos \theta'}} d\theta'$$

$$\vec{A}(\vec{r}) = -\hat{y} \frac{\mu_0 \sigma R^3 \omega \sin \psi}{2} \int_0^\pi \frac{\cos \theta' \sin \theta'}{\sqrt{R^2 + r^2 - 2Rr \cos \theta'}} d\theta'$$

74

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?

$$\vec{A}(\vec{r}) = -\hat{y} \frac{\mu_0 \sigma R^3 \omega \sin \psi}{2} \int_0^\pi \frac{\cos \theta' \sin \theta'}{\sqrt{R^2 + r^2 - 2Rr \cos \theta'}} d\theta'$$

75

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?

$$\vec{A}(\vec{r}) = -\hat{y} \frac{\mu_0 \sigma R^3 \omega \sin \psi}{2} \int_0^\pi \frac{\cos \theta' \sin \theta'}{\sqrt{R^2 + r^2 - 2Rr \cos \theta'}} d\theta'$$

$u = \cos \theta' \quad du = -\sin \theta' d\theta'$

76

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?

$$\bar{\mathbf{A}}(\vec{r}) = -\hat{\mathbf{y}} \frac{\mu_0 \sigma R^3 \omega \sin \psi}{2} \int_0^\pi \frac{\cos \theta' \sin \theta'}{\sqrt{R^2 + r^2 - 2 R r \cos \theta'}} d\theta'$$

$$u = \cos \theta' \quad du = -\sin \theta' d\theta'$$

$$\int_{-1}^{+1} \frac{u}{\sqrt{R^2 + r^2 - 2 R r u}} du$$

77

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?

$$\bar{\mathbf{A}}(\vec{r}) = -\hat{\mathbf{y}} \frac{\mu_0 \sigma R^3 \omega \sin \psi}{2} \int_0^\pi \frac{\cos \theta' \sin \theta'}{\sqrt{R^2 + r^2 - 2 R r \cos \theta'}} d\theta'$$

$$u = \cos \theta' \quad du = -\sin \theta' d\theta'$$

$$\int_{-1}^{+1} \frac{u}{\sqrt{R^2 + r^2 - 2 R r u}} du = - \frac{(R^2 + r^2 + 2 R r u)}{3 R^2 r^2} \sqrt{R^2 + r^2 - 2 R r u} \Big|_{-1}^{+1}$$

78

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?

$$\bar{\mathbf{A}}(\vec{r}) = -\hat{\mathbf{y}} \frac{\mu_0 \sigma R^3 \omega \sin \psi}{2} \int_0^\pi \frac{\cos \theta' \sin \theta'}{\sqrt{R^2 + r^2 - 2 R r \cos \theta'}} d\theta'$$

$$u = \cos \theta' \quad du = -\sin \theta' d\theta'$$

$$\int_{-1}^{+1} \frac{u}{\sqrt{R^2 + r^2 - 2 R r u}} du = - \frac{(R^2 + r^2 + 2 R r u)}{3 R^2 r^2} \sqrt{R^2 + r^2 - 2 R r u} \Big|_{-1}^{+1}$$

$$= - \frac{1}{3 R^2 r^2} [(R^2 + r^2 + 2 R r) |R - r| - (R^2 + r^2 - 2 R r) (R + r)]$$

79

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?

$$\bar{\mathbf{A}}(\vec{r}) = -\hat{\mathbf{y}} \frac{\mu_0 \sigma R^3 \omega \sin \psi}{2} \int_0^\pi \frac{\cos \theta' \sin \theta'}{\sqrt{R^2 + r^2 - 2 R r \cos \theta'}} d\theta'$$

$$u = \cos \theta' \quad du = -\sin \theta' d\theta'$$

$$\int_{-1}^{+1} \frac{u}{\sqrt{R^2 + r^2 - 2 R r u}} du = - \frac{(R^2 + r^2 + 2 R r u)}{3 R^2 r^2} \sqrt{R^2 + r^2 - 2 R r u} \Big|_{-1}^{+1}$$

$$= - \frac{1}{3 R^2 r^2} [(R^2 + r^2 + 2 R r) |R - r| - (R^2 + r^2 - 2 R r) (R + r)]$$

$$= - \frac{1}{3 R^2 r^2} [(R^2 + r^2 + 2 R r) (R - r) - (R^2 + r^2 - 2 R r) (R + r)] (R > r)$$

$$= - \frac{1}{3 R^2 r^2} [(R^2 + r^2 + 2 R r) (-R + r) - (R^2 + r^2 - 2 R r) (R + r)] (R < r)$$

$$= \begin{cases} 2r/3R^2 & \text{inside } (R > r) \\ 2R/3r^2 & \text{outside } (R < r) \end{cases}$$

80

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?

$$\begin{aligned}\bar{\mathbf{A}}(\vec{r}) &= -\hat{\mathbf{y}} \frac{\mu_0 \sigma R^3 \omega \sin \psi}{2} \int_0^\pi \frac{\cos \theta' \sin \theta'}{\sqrt{R^2 + r^2 - 2 R r \cos \theta'}} d\theta' \\ u = \cos \theta' \quad du &= -\sin \theta' d\theta' \\ \int_{-1}^{+1} \frac{u}{\sqrt{R^2 + r^2 - 2 R r u}} du &= -\frac{(R^2 + r^2 + 2 R r u)}{3 R^2 r^2} \sqrt{R^2 + r^2 - 2 R r u} \Big|_{-1}^{+1} \\ &= -\frac{1}{3 R^2 r^2} [(R^2 + r^2 + 2 R r) |R - r| - (R^2 + r^2 - 2 R r) (R + r)] \\ &= \begin{cases} 2r/3R^2 & \text{inside } (R > r) \\ 2R/3r^2 & \text{outside } (R < r) \end{cases}\end{aligned}$$

81

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?

$$\begin{aligned}\bar{\mathbf{A}}(\vec{r}) &= -\hat{\mathbf{y}} \frac{\mu_0 \sigma R^3 \omega \sin \psi}{2} \int_0^\pi \frac{\cos \theta' \sin \theta'}{\sqrt{R^2 + r^2 - 2 R r \cos \theta'}} d\theta' \\ u = \cos \theta' \quad du &= -\sin \theta' d\theta' \\ \int_{-1}^{+1} \frac{u}{\sqrt{R^2 + r^2 - 2 R r u}} du &= -\frac{(R^2 + r^2 + 2 R r u)}{3 R^2 r^2} \sqrt{R^2 + r^2 - 2 R r u} \Big|_{-1}^{+1} \\ &= -\frac{1}{3 R^2 r^2} [(R^2 + r^2 + 2 R r) |R - r| - (R^2 + r^2 - 2 R r) (R + r)] \\ &= \begin{cases} 2r/3R^2 & \text{inside } (R > r) \\ 2R/3r^2 & \text{outside } (R < r) \end{cases}\end{aligned}$$

$$\bar{\mathbf{A}}(\vec{r}) =$$

82

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?

$$\begin{aligned}\bar{\mathbf{A}}(\vec{r}) &= -\hat{\mathbf{y}} \frac{\mu_0 \sigma R^3 \omega \sin \psi}{2} \int_0^\pi \frac{\cos \theta' \sin \theta'}{\sqrt{R^2 + r^2 - 2 R r \cos \theta'}} d\theta' \\ u = \cos \theta' \quad du &= -\sin \theta' d\theta' \\ \int_{-1}^{+1} \frac{u}{\sqrt{R^2 + r^2 - 2 R r u}} du &= -\frac{(R^2 + r^2 + 2 R r u)}{3 R^2 r^2} \sqrt{R^2 + r^2 - 2 R r u} \Big|_{-1}^{+1} \\ &= -\frac{1}{3 R^2 r^2} [(R^2 + r^2 + 2 R r) |R - r| - (R^2 + r^2 - 2 R r) (R + r)] \\ &= \begin{cases} 2r/3R^2 & \text{inside } (R > r) \\ 2R/3r^2 & \text{outside } (R < r) \end{cases} \\ \bar{\mathbf{A}}(\vec{r}) &= \begin{cases} -\frac{\mu_0 \sigma R}{3} \omega r \sin \psi \hat{\mathbf{y}} & \text{inside } (R > r) \\ \end{cases}\end{aligned}$$

83

Example - 1

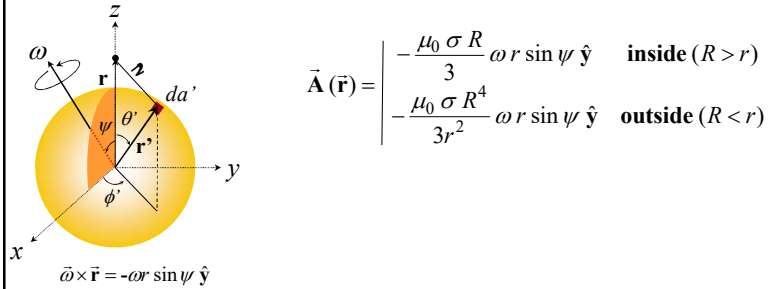
Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?

$$\begin{aligned}\bar{\mathbf{A}}(\vec{r}) &= -\hat{\mathbf{y}} \frac{\mu_0 \sigma R^3 \omega \sin \psi}{2} \int_0^\pi \frac{\cos \theta' \sin \theta'}{\sqrt{R^2 + r^2 - 2 R r \cos \theta'}} d\theta' \\ u = \cos \theta' \quad du &= -\sin \theta' d\theta' \\ \int_{-1}^{+1} \frac{u}{\sqrt{R^2 + r^2 - 2 R r u}} du &= -\frac{(R^2 + r^2 + 2 R r u)}{3 R^2 r^2} \sqrt{R^2 + r^2 - 2 R r u} \Big|_{-1}^{+1} \\ &= -\frac{1}{3 R^2 r^2} [(R^2 + r^2 + 2 R r) |R - r| - (R^2 + r^2 - 2 R r) (R + r)] \\ &= \begin{cases} 2r/3R^2 & \text{inside } (R > r) \\ 2R/3r^2 & \text{outside } (R < r) \end{cases} \\ \bar{\mathbf{A}}(\vec{r}) &= \begin{cases} -\frac{\mu_0 \sigma R}{3} \omega r \sin \psi \hat{\mathbf{y}} & \text{inside } (R > r) \\ -\frac{\mu_0 \sigma R^4}{3r^2} \omega r \sin \psi \hat{\mathbf{y}} & \text{outside } (R < r) \end{cases}\end{aligned}$$

84

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?

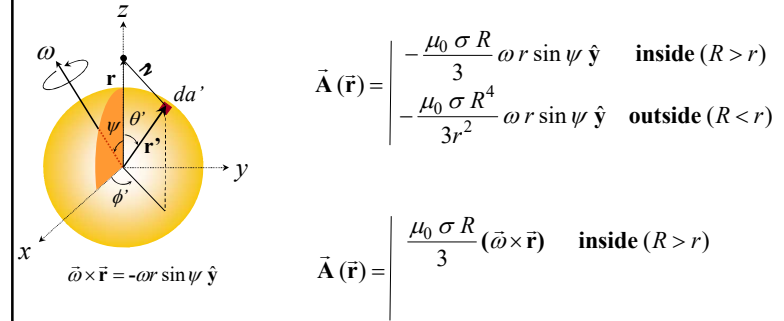


$$\vec{A}(\vec{r}) = \begin{cases} -\frac{\mu_0 \sigma R}{3} \omega r \sin \psi \hat{y} & \text{inside } (R > r) \\ -\frac{\mu_0 \sigma R^4}{3r^2} \omega r \sin \psi \hat{y} & \text{outside } (R < r) \end{cases}$$

85

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



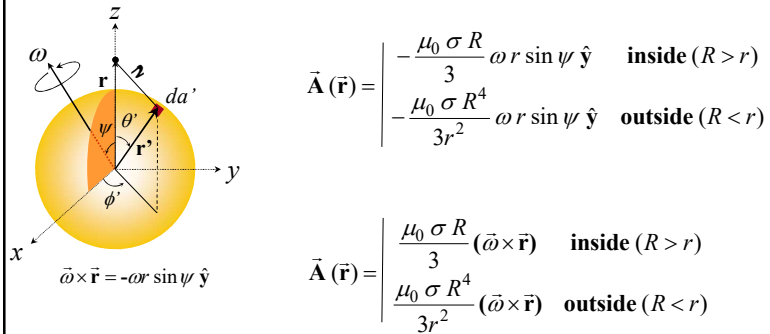
$$\vec{A}(\vec{r}) = \begin{cases} -\frac{\mu_0 \sigma R}{3} \omega r \sin \psi \hat{y} & \text{inside } (R > r) \\ -\frac{\mu_0 \sigma R^4}{3r^2} \omega r \sin \psi \hat{y} & \text{outside } (R < r) \end{cases}$$

$$\vec{A}(\vec{r}) = \begin{cases} \frac{\mu_0 \sigma R}{3} (\vec{\omega} \times \vec{r}) & \text{inside } (R > r) \end{cases}$$

86

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



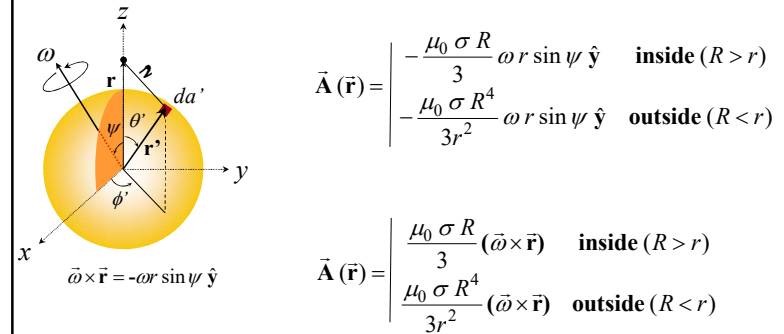
$$\vec{A}(\vec{r}) = \begin{cases} -\frac{\mu_0 \sigma R}{3} \omega r \sin \psi \hat{y} & \text{inside } (R > r) \\ -\frac{\mu_0 \sigma R^4}{3r^2} \omega r \sin \psi \hat{y} & \text{outside } (R < r) \end{cases}$$

$$\vec{A}(\vec{r}) = \begin{cases} \frac{\mu_0 \sigma R}{3} (\vec{\omega} \times \vec{r}) & \text{inside } (R > r) \\ \frac{\mu_0 \sigma R^4}{3r^2} (\vec{\omega} \times \vec{r}) & \text{outside } (R < r) \end{cases}$$

87

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



$$\vec{A}(\vec{r}) = \begin{cases} -\frac{\mu_0 \sigma R}{3} \omega r \sin \psi \hat{y} & \text{inside } (R > r) \\ -\frac{\mu_0 \sigma R^4}{3r^2} \omega r \sin \psi \hat{y} & \text{outside } (R < r) \end{cases}$$

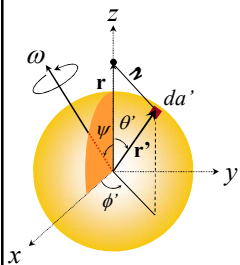
$$\vec{A}(\vec{r}) = \begin{cases} \frac{\mu_0 \sigma R}{3} (\vec{\omega} \times \vec{r}) & \text{inside } (R > r) \\ \frac{\mu_0 \sigma R^4}{3r^2} (\vec{\omega} \times \vec{r}) & \text{outside } (R < r) \end{cases}$$

No reference of the coordinate system – General Solution

88

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?

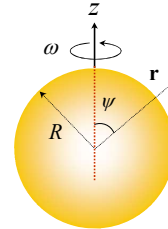


$$\vec{A}(\vec{r}) = \begin{cases} \frac{\mu_0 \sigma R}{3} (\vec{\omega} \times \vec{r}) & \text{inside } (R > r) \\ \frac{\mu_0 \sigma R^4}{3r^3} (\vec{\omega} \times \vec{r}) & \text{outside } (R < r) \end{cases}$$

89

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



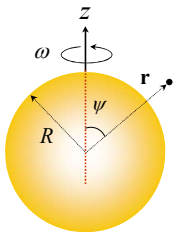
$$\vec{A}(\vec{r}) = \begin{cases} \frac{\mu_0 \sigma R}{3} (\vec{\omega} \times \vec{r}) & \text{inside } (R > r) \\ \frac{\mu_0 \sigma R^4}{3r^3} (\vec{\omega} \times \vec{r}) & \text{outside } (R < r) \end{cases}$$

Returning back to the original configuration, where ω coincides with the z-axis

90

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



$$\vec{A}(\vec{r}) = \begin{cases} \frac{\mu_0 \sigma R}{3} (\vec{\omega} \times \vec{r}) & \text{inside } (R > r) \\ \frac{\mu_0 \sigma R^4}{3r^3} (\vec{\omega} \times \vec{r}) & \text{outside } (R < r) \end{cases}$$

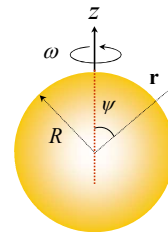
Returning back to the original configuration, where ω coincides with the z-axis

$$\vec{A}(\vec{r}) =$$

91

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



$$\vec{A}(\vec{r}) = \begin{cases} \frac{\mu_0 \sigma R}{3} (\vec{\omega} \times \vec{r}) & \text{inside } (R > r) \\ \frac{\mu_0 \sigma R^4}{3r^3} (\vec{\omega} \times \vec{r}) & \text{outside } (R < r) \end{cases}$$

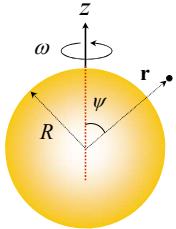
Returning back to the original configuration, where ω coincides with the z-axis

$$\vec{A}(\vec{r}) = \begin{cases} \frac{\mu_0 \sigma R \omega}{3} r \sin \psi \hat{\phi} & \text{inside } (R > r) \end{cases}$$

92

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



$$\vec{A}(\vec{r}) = \begin{cases} \frac{\mu_0 \sigma R}{3} (\vec{\omega} \times \vec{r}) & \text{inside } (R > r) \\ \frac{\mu_0 \sigma R^4}{3r^3} (\vec{\omega} \times \vec{r}) & \text{outside } (R < r) \end{cases}$$

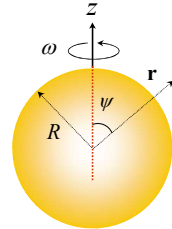
Returning back to the original configuration, where ω coincides with the z-axis

$$\vec{A}(\vec{r}) = \begin{cases} \frac{\mu_0 \sigma R \omega}{3} r \sin \psi \hat{\phi} & \text{inside } (R > r) \\ \frac{\mu_0 \sigma R^4 \omega}{3r^2} \sin \psi \hat{\phi} & \text{outside } (R < r) \end{cases}$$

93

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



$$\vec{A}(\vec{r}) = \begin{cases} \frac{\mu_0 \sigma R}{3} (\vec{\omega} \times \vec{r}) & \text{inside } (R > r) \\ \frac{\mu_0 \sigma R^4}{3r^3} (\vec{\omega} \times \vec{r}) & \text{outside } (R < r) \end{cases}$$

Returning back to the original configuration, where ω coincides with the z-axis

$$\vec{A}(\vec{r}) = \begin{cases} \frac{\mu_0 \sigma R \omega}{3} r \sin \psi \hat{\phi} & \text{inside } (R > r) \\ \frac{\mu_0 \sigma R^4 \omega}{3r^2} \sin \psi \hat{\phi} & \text{outside } (R < r) \end{cases}$$

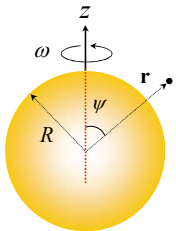
Inside the sphere

$\vec{B}$

94

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



$$\vec{A}(\vec{r}) = \begin{cases} \frac{\mu_0 \sigma R}{3} (\vec{\omega} \times \vec{r}) & \text{inside } (R > r) \\ \frac{\mu_0 \sigma R^4}{3r^3} (\vec{\omega} \times \vec{r}) & \text{outside } (R < r) \end{cases}$$

Returning back to the original configuration, where ω coincides with the z-axis

$$\vec{A}(\vec{r}) = \begin{cases} \frac{\mu_0 \sigma R \omega}{3} r \sin \psi \hat{\phi} & \text{inside } (R > r) \\ \frac{\mu_0 \sigma R^4 \omega}{3r^2} \sin \psi \hat{\phi} & \text{outside } (R < r) \end{cases}$$

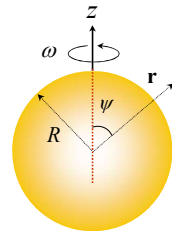
Inside the sphere

$$\vec{B} = \vec{\nabla} \times \vec{A}$$

95

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



$$\vec{A}(\vec{r}) = \begin{cases} \frac{\mu_0 \sigma R}{3} (\vec{\omega} \times \vec{r}) & \text{inside } (R > r) \\ \frac{\mu_0 \sigma R^4}{3r^3} (\vec{\omega} \times \vec{r}) & \text{outside } (R < r) \end{cases}$$

Returning back to the original configuration, where ω coincides with the z-axis

$$\vec{A}(\vec{r}) = \begin{cases} \frac{\mu_0 \sigma R \omega}{3} r \sin \psi \hat{\phi} & \text{inside } (R > r) \\ \frac{\mu_0 \sigma R^4 \omega}{3r^2} \sin \psi \hat{\phi} & \text{outside } (R < r) \end{cases}$$

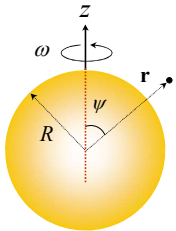
Inside the sphere

$$\vec{B} = \vec{\nabla} \times \vec{A} = \frac{2\mu_0 \sigma R \omega}{3} (\cos \theta \hat{r} - \sin \theta \hat{\theta})$$

96

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



$$\vec{A}(\vec{r}) = \begin{cases} \frac{\mu_0 \sigma R}{3} (\vec{\omega} \times \vec{r}) & \text{inside } (R > r) \\ \frac{\mu_0 \sigma R^4}{3r^3} (\vec{\omega} \times \vec{r}) & \text{outside } (R < r) \end{cases}$$

Returning back to the original configuration, where ω coincides with the z-axis

$$\vec{A}(\vec{r}) = \begin{cases} \frac{\mu_0 \sigma R \omega}{3} r \sin \psi \hat{\phi} & \text{inside } (R > r) \\ \frac{\mu_0 \sigma R^4 \omega}{3r^2} \sin \psi \hat{\phi} & \text{outside } (R < r) \end{cases}$$

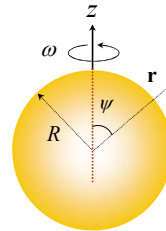
Inside the sphere

$$\vec{B} = \vec{\nabla} \times \vec{A} = \frac{2\mu_0 \sigma R \omega}{3} (\cos \theta \hat{r} - \sin \theta \hat{\theta}) = \frac{2}{3} \mu_0 \sigma R \omega \hat{z}$$

97

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



$$\vec{A}(\vec{r}) = \begin{cases} \frac{\mu_0 \sigma R}{3} (\vec{\omega} \times \vec{r}) & \text{inside } (R > r) \\ \frac{\mu_0 \sigma R^4}{3r^3} (\vec{\omega} \times \vec{r}) & \text{outside } (R < r) \end{cases}$$

Returning back to the original configuration, where ω coincides with the z-axis

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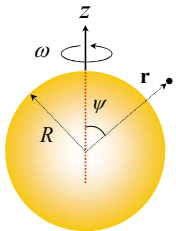
Inside the sphere

$$\vec{B} = \vec{\nabla} \times \vec{A} = \frac{2\mu_0 \sigma R \omega}{3} (\cos \theta \hat{r} - \sin \theta \hat{\theta}) = \frac{2}{3} \mu_0 \sigma R \omega \hat{z} = \frac{2}{3} \mu_0 \sigma R \vec{\omega}$$

98

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



$$\vec{A}(\vec{r}) = \begin{cases} \frac{\mu_0 \sigma R}{3} (\vec{\omega} \times \vec{r}) & \text{inside } (R > r) \\ \frac{\mu_0 \sigma R^4}{3r^3} (\vec{\omega} \times \vec{r}) & \text{outside } (R < r) \end{cases}$$

Returning back to the original configuration, where ω coincides with the z-axis

$$\vec{A}(\vec{r}) = \begin{cases} \frac{\mu_0 \sigma R \omega}{3} r \sin \psi \hat{\phi} & \text{inside } (R > r) \\ \frac{\mu_0 \sigma R^4 \omega}{3r^2} \sin \psi \hat{\phi} & \text{outside } (R < r) \end{cases}$$

Inside the sphere

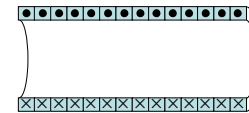
$$\vec{B} = \vec{\nabla} \times \vec{A} = \frac{2\mu_0 \sigma R \omega}{3} (\cos \theta \hat{r} - \sin \theta \hat{\theta}) = \frac{2}{3} \mu_0 \sigma R \omega \hat{z} = \frac{2}{3} \mu_0 \sigma R \vec{\omega}$$

Field is Uniform inside

99

Important concepts - 2**Magnetic Vector Potential**

Vector potential of an infinite solenoid, radius R with n turns per unit length and current I ?



We can not use $\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{J}(\vec{r}')}{r} d\tau'$ $\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{I}}{r} dl'$

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{K}}{r} da'$$

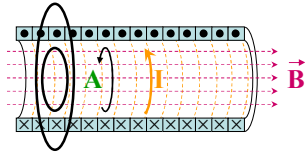
Cause current extends to infinity !

100

Important concepts - 2

Magnetic Vector Potential

Vector potential of an infinite solenoid, radius R with n turns per unit length and current I ?



$$\oint \vec{A} \cdot d\vec{l} = \int (\vec{\nabla} \times \vec{A}) \cdot d\vec{a} = \int \vec{B} \cdot d\vec{a}$$

Inside $\oint \vec{A} \cdot d\vec{l} = A(2\pi r) = \int \vec{B} \cdot d\vec{a} = \mu_0 n I (\pi r^2)$ $\vec{A} = \frac{\mu_0 n I}{2} r \hat{\phi} \quad r < R$

Outside $A(2\pi R) = \int \vec{B} \cdot d\vec{a} = \mu_0 n I (\pi R^2)$ $\vec{A} = \frac{\mu_0 n I}{2} \frac{R^2}{r} \hat{\phi} \quad r > R$

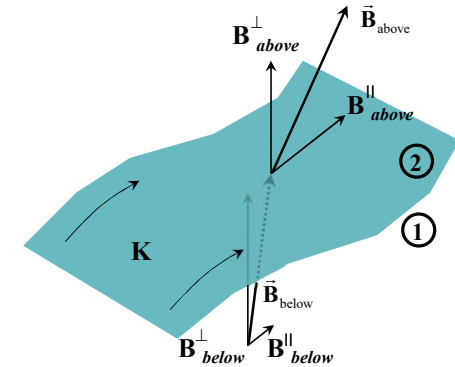
Verify by calculating $\text{curl } \vec{A} = \vec{B}$? $\text{div } \vec{A} = 0$?

101

Magnetostatic Boundary Conditions

Electric field suffers discontinuity at the surface charge

Magnetic field suffers discontinuity at the surface current



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Magnetostatic Boundary Conditions

Electric field suffers discontinuity at the surface charge

Magnetic field suffers discontinuity at the surface current

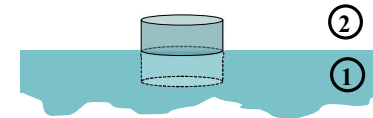


103

Magnetostatic Boundary Conditions

Electric field suffers discontinuity at the surface charge

Magnetic field suffers discontinuity at the surface current



104

Magnetostatic Boundary Conditions

Electric field suffers discontinuity at the surface charge

Magnetic field suffers discontinuity at the surface current

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\int_V (\vec{\nabla} \cdot \vec{B}) d\tau = \oint_S \vec{B} \cdot d\vec{a} = \vec{B}_2 \cdot \hat{n}_2 \Delta a + \vec{B}_1 \cdot \hat{n}_1 \Delta a + w$$

$$= \vec{B}_2 \cdot \hat{n} \Delta a - \vec{B}_1 \cdot \hat{n} \Delta a + w = \hat{n} \cdot (\vec{B}_2 - \vec{B}_1) \Delta a + w = 0$$

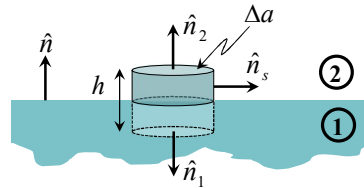
$$\lim_{h \rightarrow 0} \hat{n} \cdot (\vec{B}_2 - \vec{B}_1) \Delta a + \lim_{h \rightarrow 0} w = 0$$



$$\hat{n} \cdot (\vec{B}_2 - \vec{B}_1) = 0$$

$$B_{\text{above}}^{\perp} = B_{\text{below}}^{\perp}$$

Normal comp of B



105

Magnetostatic Boundary Conditions

Electric field suffers discontinuity at the surface charge

Magnetic field suffers discontinuity at the surface current

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} \rightarrow \int_S (\vec{\nabla} \times \vec{B}) \cdot d\vec{a} = \mu_0 \int_S \vec{J} \cdot d\vec{a} \rightarrow \oint_C \vec{B} \cdot d\vec{l} = \mu_0 \int_S \vec{J} \cdot d\vec{a}$$

$$\vec{B}_2 \cdot \hat{t}_2 \Delta l + \vec{B}_1 \cdot \hat{t}_1 \Delta l + w = \vec{B}_2 \cdot \hat{t} \Delta l - \vec{B}_1 \cdot \hat{t} \Delta l + w = \hat{t} \cdot (\vec{B}_2 - \vec{B}_1) \Delta l + w$$

$$= (\hat{n}' \times \hat{n}) \cdot (\vec{B}_2 - \vec{B}_1) \Delta l + w = \hat{n}' \cdot \hat{n} \times (\vec{B}_2 - \vec{B}_1) \Delta l + w = \mu_0 \vec{J} \cdot \hat{n}' h \Delta l$$

$$\lim_{h \rightarrow 0} (\hat{n}' \cdot \hat{n} \times (\vec{B}_2 - \vec{B}_1) \Delta l + w) = \mu_0 \lim_{h \rightarrow 0} \vec{J} \cdot \hat{n}' h \Delta l$$

$$\hat{n}' \cdot [\hat{n} \times (\vec{B}_2 - \vec{B}_1)] = \mu_0 \hat{n}' \cdot \lim_{h \rightarrow 0} (h \vec{J}) = \mu_0 \hat{n}' \cdot \vec{K}$$

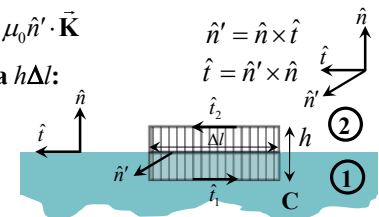
Total current ΔI through the area hΔl:

$$\Delta I = J h \Delta l$$

$$\lim_{h \rightarrow 0} \Delta I = \lim_{h \rightarrow 0} (J h \Delta l) = K \Delta l$$



$$\hat{n} \times (\vec{B}_2 - \vec{B}_1) = \mu_0 \vec{K}$$



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Magnetostatic Boundary Conditions

Electric field suffers discontinuity at the surface charge

Magnetic field suffers discontinuity at the surface current

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} \rightarrow \hat{n} \times (\vec{B}_2 - \vec{B}_1) = \mu_0 \vec{K}$$

$$\hat{n} \times (\vec{B}_2^{\parallel} - \vec{B}_1^{\parallel}) = \mu_0 \vec{K}$$

$$\vec{B}_2^{\parallel} - \vec{B}_1^{\parallel} = \mu_0 (\vec{K} \times \hat{n})$$

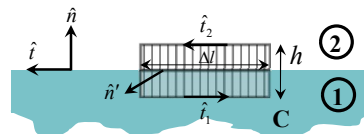
$$\vec{B} = \vec{B}^{\perp} + \vec{B}^{\parallel} = B^{\perp} \hat{n} + \vec{B}^{\parallel}$$

$$\hat{n} \times \vec{B} = \hat{n} \times (B^{\perp} \hat{n} + \vec{B}^{\parallel}) = \hat{n} \times \vec{B}^{\parallel}$$

$$(\hat{n} \times \vec{B}) \times \hat{n} = -\hat{n} \times (\hat{n} \times \vec{B})$$

$$= -[\hat{n}(\hat{n} \cdot \vec{B}) - \vec{B}(\hat{n} \cdot \hat{n})]$$

$$= \vec{B} - B^{\perp} \hat{n} = \vec{B}^{\parallel}$$



107

BC - Summary

$$B_{\text{above}}^{\perp} = B_{\text{below}}^{\perp}$$

$$E_{\text{above}}^{\perp} - E_{\text{below}}^{\perp} = \frac{1}{\epsilon_0} \sigma$$

$$B_{\text{above}}^{\parallel} - B_{\text{below}}^{\parallel} = \mu_0 K$$

$$E_{\text{above}}^{\parallel} = E_{\text{below}}^{\parallel}$$

$$B_{\text{above}}^{\parallel} = B_{\text{below}}^{\parallel}$$

$$\vec{B}_{\text{above}} - \vec{B}_{\text{below}} = \mu_0 (\vec{K} \times \hat{n})$$

$$\vec{E}_{\text{above}} - \vec{E}_{\text{below}} = \frac{\sigma}{\epsilon_0} \hat{n}$$

$$\vec{A}_{\text{above}} = \vec{A}_{\text{below}}$$

$$V_{\text{above}} = V_{\text{below}}$$

$$\frac{\partial \vec{A}_{\text{above}}}{\partial n} - \frac{\partial \vec{A}_{\text{below}}}{\partial n} = -\mu_0 \vec{K}$$

$$\frac{\partial V_{\text{above}}}{\partial n} - \frac{\partial V_{\text{below}}}{\partial n} = -\frac{1}{\epsilon_0} \sigma$$

108

Multipole expansion of vector potential

$$\vec{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{J}(\mathbf{r}')}{r} d\tau'$$

From cosine law:

$$r = (r^2 + r'^2 - 2rr' \cos \theta)^{1/2}$$

$$= r \left(1 + \frac{r'^2}{r^2} - \frac{2r'}{r} \cos \theta \right)^{1/2}$$

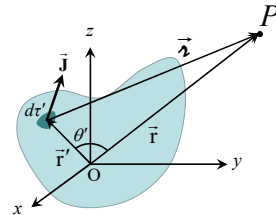
$$\approx r \left(1 - \frac{2r'}{r} \cos \theta \right)^{1/2} \quad r \gg r'$$

$$\frac{1}{r} = \frac{1}{r} \left(1 - \frac{2r'}{r} \cos \theta \right)^{-1/2}$$

$$\approx \frac{1}{r} \left(1 + \frac{r'}{r} \cos \theta \right)$$

$$\vec{A}(\mathbf{r}) = \frac{\mu_0}{4\pi r} \int \vec{J}(\mathbf{r}') d\tau' \quad \text{monopole}$$

$$+ \frac{\mu_0}{4\pi r^2} \int (\hat{\mathbf{r}} \cdot \mathbf{r}') \vec{J}(\mathbf{r}') d\tau' + \dots \quad \text{dipole}$$



109

Multipole expansion of vector potential

Current Carrying loop

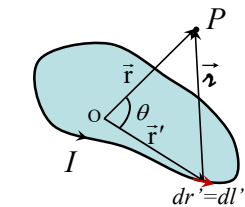
$$\vec{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \oint \frac{\mathbf{I} d\vec{l}'}{r}$$

$$= \underbrace{\frac{\mu_0 I}{4\pi r} \oint d\vec{l}'}_{\text{monopole}} + \underbrace{\frac{\mu_0 I}{4\pi r^2} \oint r' \cos \theta d\vec{l}'}_{\text{dipole}} \dots$$

$$\vec{A}_{\text{dip}}(\mathbf{r}) = \frac{\mu_0 I}{4\pi r^2} \oint r' \cos \theta d\vec{l}'$$

$$= \frac{\mu_0 I}{4\pi r^2} \oint (\mathbf{r}' \cdot \hat{\mathbf{r}}) d\vec{l}'$$

$$= -\frac{\mu_0 I}{4\pi r^2} \hat{\mathbf{r}} \times \int d\vec{a}' = -\frac{\mu_0}{4\pi r^2} \hat{\mathbf{r}} \times \vec{m} = \frac{\mu_0}{4\pi} \frac{\vec{m} \times \hat{\mathbf{r}}}{r^2}$$



$$\frac{1}{r} = \frac{1}{r} \left(1 + \frac{r'}{r} \cos \theta \right)$$

$$\mathbf{r}' \cdot \hat{\mathbf{r}} = r' \cos \theta$$

$$\oint (\mathbf{r}' \cdot \hat{\mathbf{r}}) d\vec{l}' = -\hat{\mathbf{r}} \times \int d\vec{a}'$$

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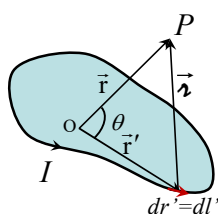
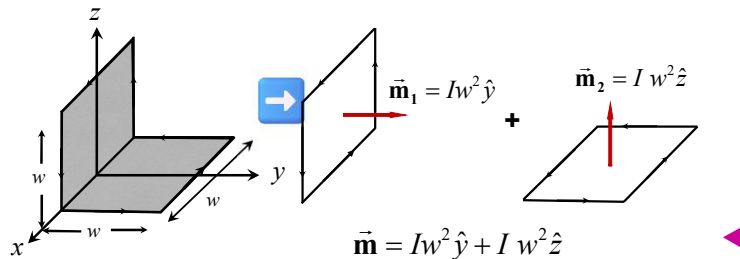
Multipole expansion of vector potential

Current Carrying loop

$$\vec{A}_{\text{dip}}(\mathbf{r}) = \frac{\mu_0}{4\pi} \frac{\vec{m} \times \hat{\mathbf{r}}}{r^2}$$

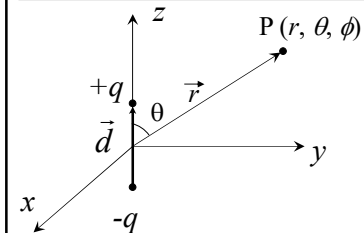
$$\vec{m} = I \int d\vec{a}' = I \vec{a}$$

Example:



111

Electric Dipole

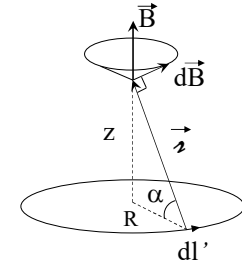


Potential

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{q\vec{d} \cdot \hat{\mathbf{r}}}{r^2}$$

Electric Field

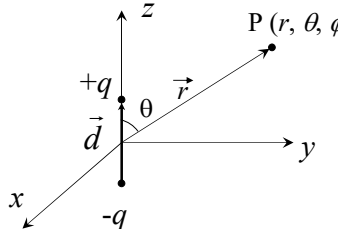
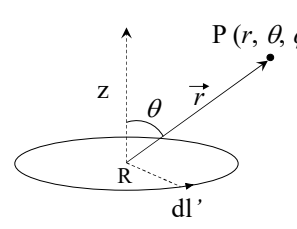
$$\vec{E}_{\text{dip}}(r, \theta) = \frac{1}{4\pi\epsilon_0} \frac{p}{r^3} \left[(2 \cos \theta) \hat{\mathbf{r}} + (\sin \theta) \hat{\boldsymbol{\theta}} \right]$$



$$\vec{B}(z) = \frac{\mu_0}{2\pi} (I \pi R^2) \frac{1}{r^3} \hat{z}$$

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Electric Dipole

Potential

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{q\vec{d} \cdot \hat{r}}{r^2}$$

Electric Field

$$\vec{E}_{dip}(r, \theta) = \frac{1}{4\pi\epsilon_0} \frac{p}{r^3} \left[(2\cos\theta)\hat{r} + (\sin\theta)\hat{\theta} \right]$$

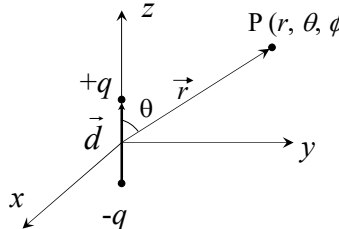
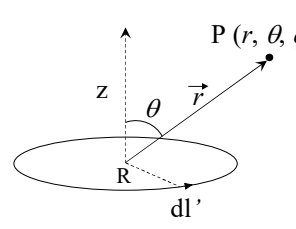
Potential $\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int I \frac{d\vec{l}'}{r}$

$$\vec{A}_{dip}(\vec{r}) = \frac{\mu_0}{4\pi} \frac{\vec{m} \times \hat{r}}{r^2}$$

$$\vec{m} = I \int d\vec{a} = I \vec{a}$$

113

Electric Dipole

Potential

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{q\vec{d} \cdot \hat{r}}{r^2}$$

Electric Field

$$\vec{E}_{dip}(r, \theta) = \frac{1}{4\pi\epsilon_0} \frac{p}{r^3} \left[(2\cos\theta)\hat{r} + (\sin\theta)\hat{\theta} \right]$$

Potential $\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int I \frac{d\vec{l}'}{r}$

$$\vec{A}_{dip}(\vec{r}) = \frac{\mu_0}{4\pi} \frac{\vec{m} \times \hat{r}}{r^2}$$

$$\vec{m} = I \int d\vec{a} = I \vec{a}$$

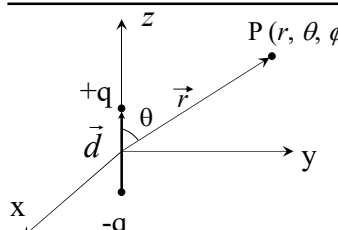
Magnetic Field $\vec{B}_{dip} = \vec{\nabla} \times \vec{A}$

$$\vec{B}_{dip}(r, \theta) = \frac{\mu_0 m}{4\pi r^3} \left[(2\cos\theta)\hat{r} + (\sin\theta)\hat{\theta} \right]$$

Dipole field

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Electric Dipole



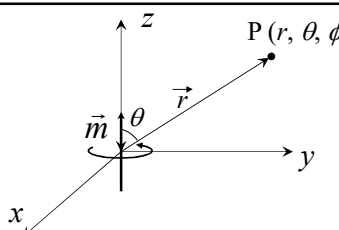
Potential

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{q\vec{d} \cdot \hat{r}}{r^2}$$

Electric Field

$$\vec{E}_{dip}(r, \theta) = \frac{1}{4\pi\epsilon_0} \frac{p}{r^3} \left[(2\cos\theta)\hat{r} + (\sin\theta)\hat{\theta} \right]$$

Magnetic Dipole



Potential $\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int I \frac{d\vec{l}'}{r}$

$$\vec{A}_{dip}(\vec{r}) = \frac{\mu_0}{4\pi} \frac{\vec{m} \times \hat{r}}{r^2}$$

$$\vec{m} = I \int d\vec{a} = I \vec{a}$$

Magnetic Field $\vec{B}_{dip} = \vec{\nabla} \times \vec{A}$

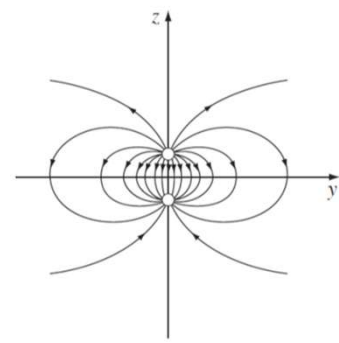
$$\vec{B}_{dip}(r, \theta) = \frac{\mu_0 m}{4\pi r^3} \left[(2\cos\theta)\hat{r} + (\sin\theta)\hat{\theta} \right]$$

Dipole field

115

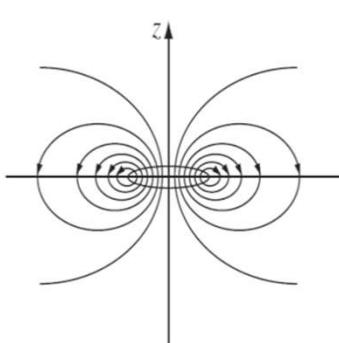
Electric Dipole

Physical



Magnetic Dipole

Physical



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